



Probability & Statistics



Outcomes

Students are expected to gain

- Knowledge about real world applications of Probability and Statistics in Machine Learning and related fields.
- Benefits of converting the knowledge and skills into job opportunities and/or career as a life long researcher or professional.



At the end of the session, you will be able to...





Contents

- Introduction
- The Law of Large numbers
- Probability and Probability Distributions
- Statistics
- Random Numbers vs. Random Variables
- Entropy
- Application Domains



Let's start with an exercise...

The Law of Large numbers



Empirical Probability vs. Theoretical Probability



- Intuitively, the probability of an event **a** could be defined as:

$$P(a) = \lim_{n \rightarrow \infty} \frac{N(a)}{n}$$

Where $N(a)$ is the number that event a happens in n trials



Joint Probability

- Joint probability is the probability of two events happening together. Joint probability can also be described as the probability of the intersection of two (or more) events.
- Dependent vs. Independent events
- You have 52 candidates for a committee. Four are persons aged 18 to 21. If you randomly select one person, and then (without replacing the first person's name), randomly select a second person, what is the probability both people will be between 18 and 21 years old?



Statistics

- is a science of data which involves
 - Collecting
 - Classifying
 - Summarizing
 - Analyzing
 - Organizing
 - Interpreting numerical information



Types of Statistics

- Descriptive Statistics
- Inferential Statistics



Probability Distributions

- ◆ Probability Density Functions
- ◆ Cumulative Distribution functions
- ◆ 5 number summary



Examples

- Tossing a coin
- Rolling a dice
- 8-bit gray scale image and its pixel intensity distribution
- Softmax function in neural network



Measures

- Measures of central tendency
 - Mean
 - Mode
 - Median
- Measures of dispersion
 - Standard Deviation
 - Variance



Measures of Center

Measures of Center

A **measure of center** is a measure that represents the center, or typical value, of a data set. The *mean*, *median*, and *mode* are measures of center.

Mean	Median	Mode
The mean of a numerical data set is the sum of the data divided by the number of data values. The symbol $\bar{x}$ represents the mean. It is read as “<i>x</i>-bar.”	The median of a numerical data set is the middle number when the values are written in numerical order. When a data set has an even number of values, the median is the mean of the two middle values.	The mode of a data set is the value or values that occur most often. There may be one mode, no mode, or more than one mode. Mode is the only measure of center that can represent a nonnumerical data set.



Measures of Center

Find the mean, median and mode of the following data set:

14 17 19 21 21 22 27 29 29 29 31

Mean = _____

Median = _____

Mode = _____

Which one seems to **best** “represent” the center of this set? Why?



Measures of Variability

Variability in statistics refers to how much the data points in a dataset vary or differ from one another. It is a measure of how spread out or dispersed the data is around the central tendency, such as the mean or median.

- Range
- Interquartile Range (IQR)
- Standard Deviation
- Variance



Measures of Variability

A dataset with high variability will have a wider range of values, while a dataset with low variability will have values that are more tightly clustered around the central tendency.



Range

Range: The range is a measure of variability in statistics that tells us the difference between the highest and lowest values in a dataset.

The range can give us an idea of the spread or dispersion of the data in a dataset, but it only considers the two extreme values and does not take into account the distribution of the other values.



Interquartile Range (IQR)

- Interquartile range (IQR) is a measure of variability in statistics that describes the spread of the middle 50% of the data in a dataset.
- It is calculated as the difference between the upper quartile (Q3) and the lower quartile (Q1) of the dataset.



Variance

- Variance is a measure of variability in statistics that describes how spread out the data in a dataset are around the mean.
- It is calculated as the average squared deviation of each data point from the mean.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

σ = population standard deviation

N = the size of the population

x_i = each value from the population

μ = the population mean



Standard Deviation

- Standard deviation is a measure of variability in statistics that describes how much the data in a dataset are spread out around the mean.
- It is calculated as the square root of the variance, which is the average of the squared deviations of each data point from the mean.



Measures of Position

Measures of position give us a way to see where a certain data point or value falls in a sample or distribution.

- Percentiles
- Quantiles
- Standard Scores

([Link](#))



5 Number Summary

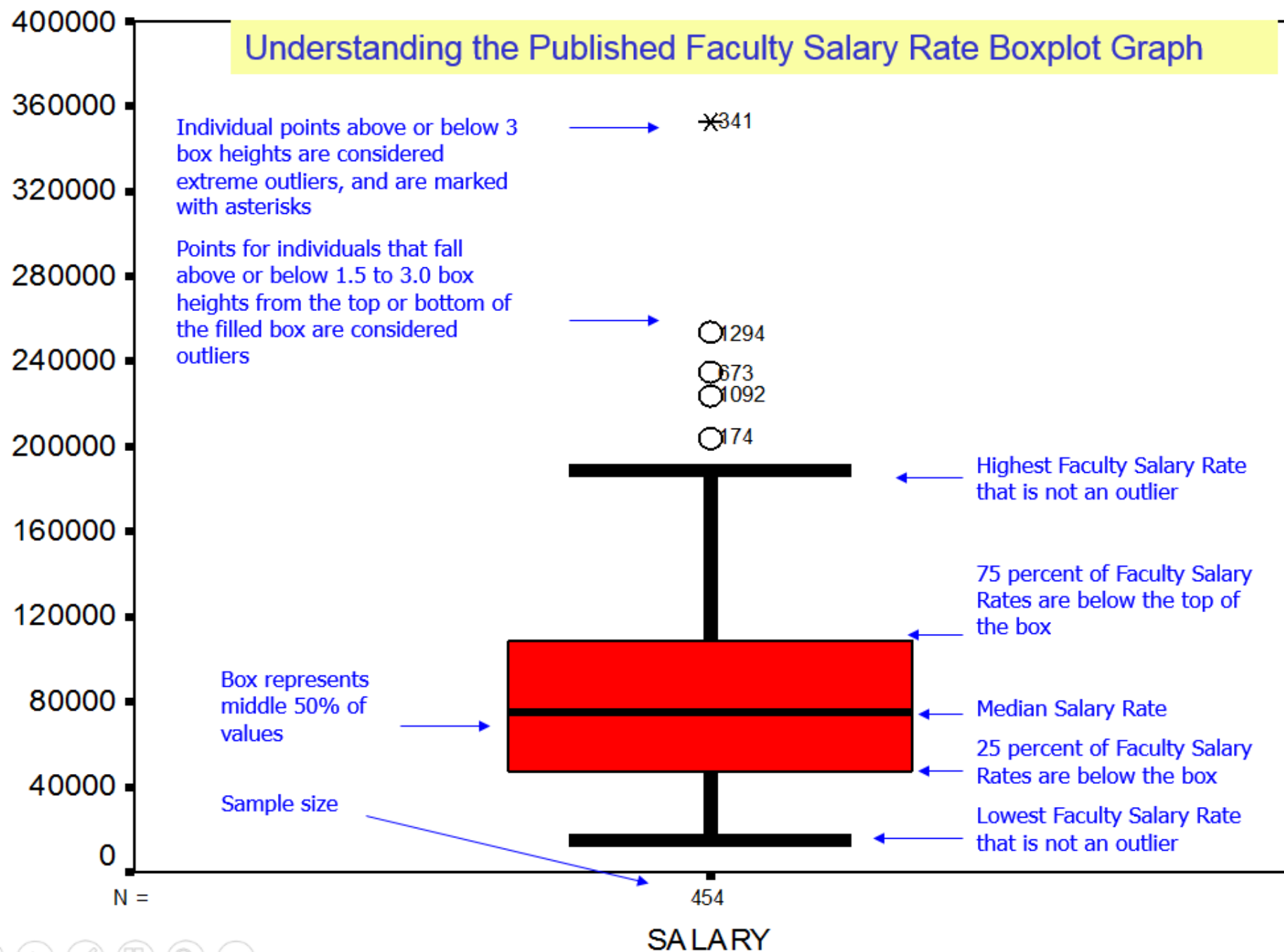
- Representation of 5-number summary of a dataset
 - Min
 - Quartile 1 (Q1)
 - Median (Q2)
 - Q3
 - Max



Visualization

- A box plot can be useful to compare two different sample distributions of the same population.
- An illustration using R

Understanding the Published Faculty Salary Rate Boxplot Graph





Example

For the following 13 real estate prices, calculate the *IQR* and determine if any prices are potential outliers. Prices are in dollars.

389,950; 230,500; 158,000; 479,000; 639,000; 114,950; 5,500,000; 387,000; 659,000; 529,000; 575,000; 488,800; 1,095,000

Answer

Order the data from smallest to largest.

114,950; 158,000; 230,500; 387,000; 389,950; 479,000; 488,800; 529,000; 575,000; 639,000; 659,000; 1,095,000; 5,500,000

$$M = 488,800$$

$$Q_1 = \frac{230,500 + 387,000}{2} = 308,750$$

$$Q_3 = \frac{639,000 + 659,000}{2} = 649,000$$

$$IQR = 649,000 - 308,750 = 340,250$$

$$(1.5)(IQR) = (1.5)(340,250) = 510,375$$

$$Q_1 - (1.5)(IQR) = 308,750 - 510,375 = -201,625$$

$$Q_3 + (1.5)(IQR) = 649,000 + 510,375 = 1,159,375$$

No house price is less than $-201,625$. However, $5,500,000$ is more than $1,159,375$. Therefore, $5,500,000$ is a potential **outlier**.



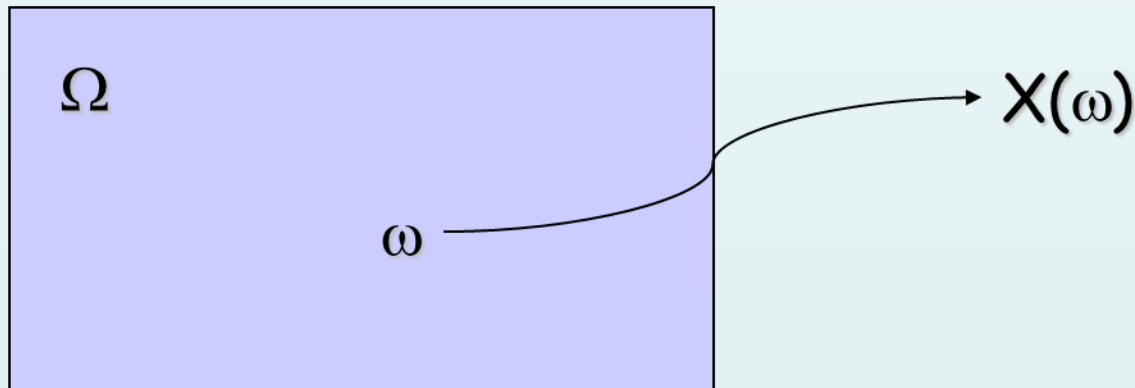
Random Numbers





Random Variables

- A (scalar) random variable X is a function that maps the outcome of a random event into real scalar values





Types of Random Variables (R.V.)

1. Continuous Random Variable
2. Discrete Random Variable



Types of Random Variables (R.V.)

- Suppose the fire department mandates that all fire fighters must weigh between 150 and 250 pounds. The weight of a fire fighter would be an example of a continuous variable; since a fire fighter's weight could take on any value between 150 and 250 pounds.
- Suppose we flip a coin twice and count the number of heads. The number of heads?



Random Variable Distributions

- Cumulative Probability Distribution (CDF)

$$F_X(x) = P(X \leq x)$$

- Probability Density Function (PDF)

$$p_X(x) = \frac{dF_X(x)}{dx}$$



$$\int_{-\infty}^{\infty} p_X(x) dx = 1.0$$



Uniform Distribution

- A R.V. X that is uniformly distributed between x_1 and x_2 has density function:



$$p_X(x) = \begin{cases} \frac{1}{x_2 - x_1} & x_1 \leq x \leq x_2 \\ 0 & \text{otherwise} \end{cases}$$

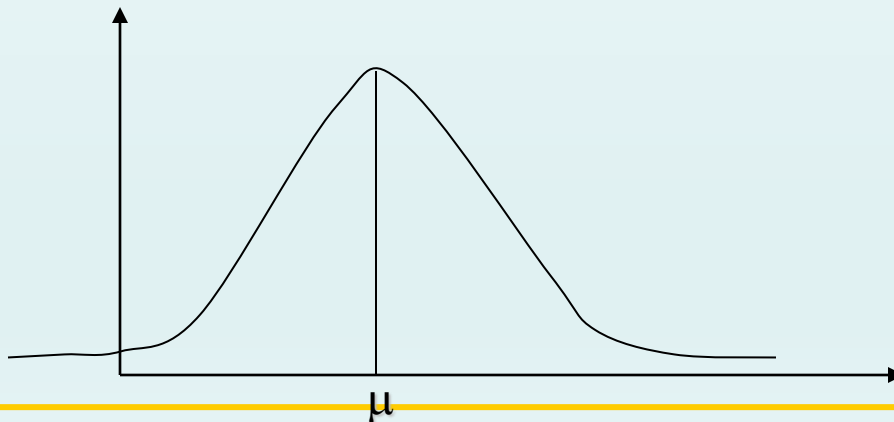


Gaussian (Normal) Distribution

- A R.V. X that is normally distributed has density function:

$$p_X(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp -\frac{(x - \mu)^2}{2\sigma^2}$$

The most
important
formula





Acknowledgement



Carl Fredrich Gauss



- Let's have another exercise....

Which of the two statements contains no information?

1. Tomorrow is an off-day.
2. Tomorrow is a working day.



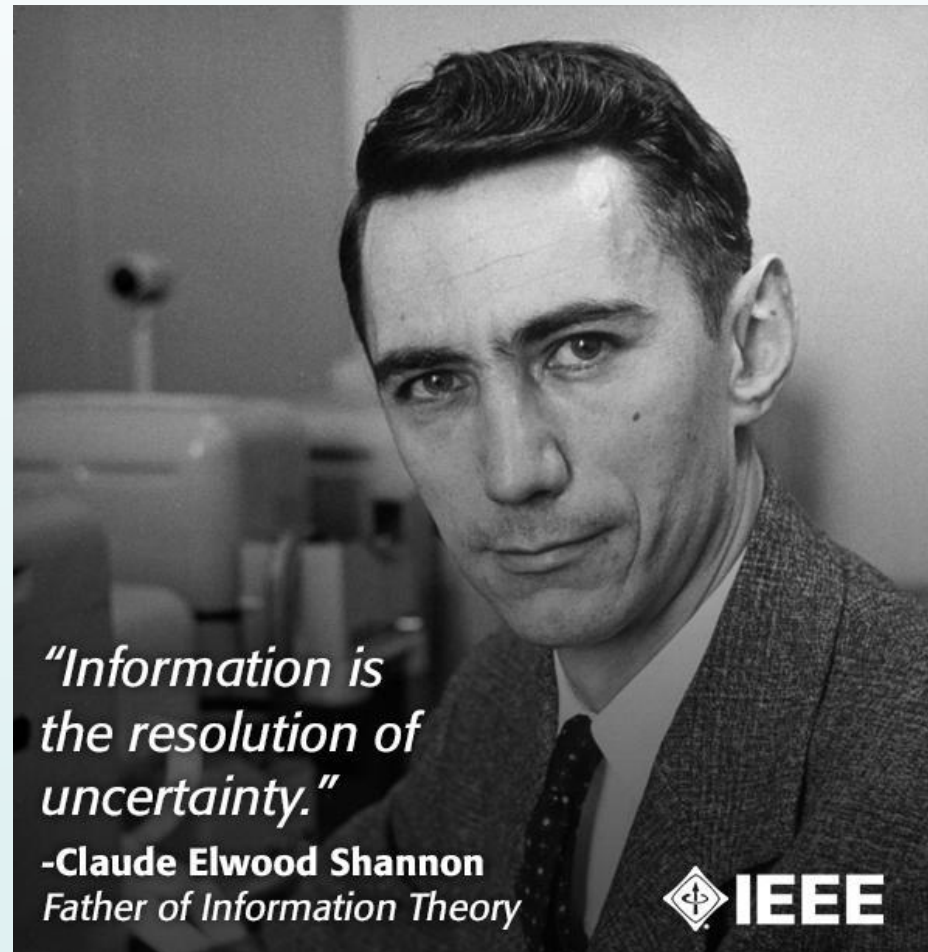
Entropy

- Entropy is a measure of uncertainty.
- The entropy for the facts which do not provide any new information is zero.
- For n equally probable possible messages, if the probability p is 1/n, then the information conveyed by a message is $-\log_2(p)$.

$$E = -\sum_{i=1}^n p_i \log p_i$$



Acknowledgement





More on Entropy

- If a given probability distribution is uniform, then its entropy is 1.
 - Tossing a fair coin
 - $P(0.5, 0.5)$
 - $E(P) = 1$
- More uniform the distribution is, the greater is its entropy.
 - $P(0.67, 0.33) = 0.92$
 - $P(1, 0) = 0$ (a biased coin)



Entropy Usage

- Classification using Decision Tree
- Information Theory



Application Domains

- Machine Learning
- Image Processing & Computer Vision
- Speech/Audio/Music Processing
- Biometric Authentication
- Recommender Systems
- Sentiment Analysis
- Video Processing
- Information Theory
- Queuing Theory
- Networks
- Cloud Computing
- Stock Market Prediction



Bayes Theorem

- Next class



Books

- A First Course on Probability by Sheldon Ross



Disclaimer

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